# Physics 468/668 (Spring 2018)
# Thermodynamics/Statistical Physics
# Mo/We 1:00-2:15 p.m., BPB 250


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| Instructor | Prof. Qiang Zhu |
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| Office | BPB 232 |
| Office hrs | Mo/We 2:15-3:15 p.m. |


## Course Outline:

- Statistical basis of thermodynamics
- Ensemble theory
- (Micro) canonical ensemble
- Grand canonical ensemble
- Phase transitions

**Prerequisite:** PHY 182, PHY 467
**Credit Hours:** 3
**Textbook:** *Statistical Mechanics*, R. K. Pathria and P. D. Beale

## Grade Distribution:

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| Assignments | 30% |
| Midterm Exam | 30% |
| Final Exam | 40% |
| Extra Credits | 10% |

## Course Description:

This course covers elements of statistical physics from fundamental of classical statistic mechanics to quantum statistics. Approximately, we spend two weeks for each chapter (1-7 in Pathria's textbook). The weekly coverage might change as it depends on the progress of the class. There will be one exam during the semester, either take-home or in-class work. This final exam will cover all materials taught in this semester. You may work with others on the homework, but take-home exams must be done **strictly by yourself**. Barring documentable emergencies or observance of a certifiable religious holiday, all exams must be taken at the time and place specified.

## Learning Outcomes:

- understand the microscopic basis for thermodynamics
- understand the macroscopic properties from micro level
- understand phase transition